

Deep CNN-Supported Ensemble CADx Architecture to Diagnose Malaria by Medical Image



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Abstract Identifying patients, infected with the virulent disease, malaria requires a reliable and quick diagnosis of blood cells. This paper presents a computer-aided diagnosing (CADx) method supported by a deep convolutional neural network (CNN) for assisting clinicians to detect malaria by medical image. We employed the VGG-19 and ResNet-50 architectures to create several models for two types of study (parasitized and uninfected erythrocytes). To enhance the model's performance, an ensemble technique was applied, followed by which, the best model selected by performance measuring metrics. Our proposed model was qualified and examined upon a standard microscopic set of images collected from the National Institute of Health (NIH). The final result was analogized with other techniques, where the accuracy of this model was 96.7% for patient-level detection. To resolute the limitations and minimizing errors regarding automated malaria detection, the proposed model proved to be an appropriate strategy for distant regions and emergencies.

Keywords Image processing · Deep learning · Ensemble · ResNet-50 · VGG-19

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1 Introduction

Malaria, a mosquito-borne blood disorder, is an endemic in many regions and an epidemic in some parts of the world. It is mainly transmitted through the female *Anopheles* mosquito bite that carries the *Plasmodium* parasites, infecting humans through showcasing symptoms of very high fevers, shivering, and illness [1]. Even though the consequences of *Plasmodium falciparum* is considered fatal, there are various types of parasites generating this disease, involving *Plasmodium malariae*, *Plasmodium ovale*, and *Plasmodium vivax* [2]. According to the World Health Organization (WHO), malaria infects more than 200 million people [3], whereas it causes more than 400,000 deaths [4] every year around the globe. The susceptibility of this type of infectious disease outbreak is depended on factors like poverty level, health-care access, governmental action, weather condition, and disease vector—a medium that carries and spreads disease to other organisms, where the fast-breeding insect, mosquitos are ill-famed for malaria and once infected, a human can also transmit this disease by sharing syringes, organ transplants, or blood transfusions [5].

The most common and reliable system to detect malaria is the infinitesimal testing of thick–thin plasma smear. In this process, the blood sample from a possible patient is placed on a slide and dyed by a divergent substance for spotting infectious red blood cells by a clinician. The accuracy of the examination primarily depends on the expertise of the medical personnel, who usually has to count 5000 red blood cells manually—an exhausting process [6]. As such, scientists developed rapid diagnosis testing (RDT), a faster technique to detect malaria, requiring a blood sample and buffer to generate a result [7]. Another approach is the polymerase chain reaction (PCR), which has a constrained efficiency [8]. For these reasons, the most effective solution is the combination of the accuracy of the microscopic testing and the speed of RTD [9].

On the other hand, employing machine learning (ML) for image analysis-based computer-aided diagnosis (CADx) software solved major issues regarding malaria detection [10–12]. Moreover, using deep learning (DL) algorithm showed better performance by overcoming the limitations of size, angle, and variability, background, and region of interest (ROI) position [13]. This paper focuses on developing a CADx technique to differentiate infected and uninfected cells from the blood smear image of a possible patient given on a microscopic lens. Firstly, a standard and large image subset of malaria were collected from the National Institute of Health (NIH), including 27,558 images—separated by two classes [14]. Then two state-of-the-art models through deep convolutional neural network (CNN) architectures (VGG-19 and ResNet-50) were created. A probability of parasitized (infected) blood cells was allocated on each view by a 19-layer CNN for visual geometry group (VGG) and 50-layer CNN for the residual neural network (ResNet), which were then averaged to estimate the final probability. Finally, an ensemble of the models was performed and compared to the classifiers for performance evaluation metrics, where the result exhibited that the system predicted the malaria risk factor with 96.7% accuracy.

The summary of this work can be drawn as follows,

- Employing a set of standard architecture to measure all parameters.
- Assembling several models to improve overall performance.
- Differentiating with notable classification methods.

The rest of the sections demonstrated the related works, proposed methodology, and experimental results.

2 Related Works

In 2018, Rajaraman et al. [2] at NIH employed six pre-trained CNN architectures, consisting of AlexNet, VGG-16, ResNet-50, Xception, DenseNet-121, and a modified DL architecture to enhance the detection methodology of malarial parasites utilizing microscopic thin blood smear images. It was a multi-step process consisting of feature extraction to classify parasitized/uninfected cells, fivefold cross-validation to decrease bias and error generalization, selecting optimal-layer to extract features and underlying data, and performance testing to identify previously trained and customized model difference. With 96 MB file size, the authors reported a 95.9% prediction accuracy of the parasitized or uninfected red blood cells (RBC) for patient-level detection, where the model processing (training and testing) time was around 24 h. Later in 2019, the authors [15] represented a technology by reducing model variance, increasing robustness, and generalization by creating a model ensemble. To prevent data leakage and reduce errors, cross-validation was performed, followed by a performance evaluation metrics. The ensemble design composed of VGG-19 and SqueezeNet seemed to outperform previous methodologies. In 2020, Fuhad et al. [16] offered a system using a range of techniques consisting of knowledge distillation, data augmentation, auto-encoder, and feature extraction via CNN standard. The model was analyzed through support vector machine (SVM) or k -nearest neighbor (k -NN), and performed under 3 training processes requiring around 4600 floating-point operations. Considering performance inference within a second; the model might be deployed for practical systems.

In 2017, Bibin et al. [17] introduced a six-layered deep belief network (DBN) for identifying the disease. The method was based on an automated decision support system, which used binary classifier to classify parasite or non-parasite cells by training a model consisting of 4100 images. The pre-training of the modeling was completed through accumulating confined Boltzmann machines employing a contrastive alteration process. To train, image feature extraction and the model's visible difference initialization were done, where color-texture was employed as a feature vector. Lastly, the model was fine-tuned through a backpropagation algorithm followed by optimization. In 2018, Devi et al. [18] developed a system on histogram-based feature for malaria identification. The erythrocyte (RBC) classification system created by preprocessing, segmentation, and feature extraction from different color channels. The color set of the histogram feature included saturation, green channel, chrominance channel, and absolute difference histogram of red-green channels from

red–green–blue (RGB) image. The optimal features of 36 features for each classifier selected by evaluating feature combinations; for the classifiers: artificial neural networks (ANN), SVM, k -NN, and Naive Bayes. The ANN distinguisher provided highest detection rate.

The inefficiency of the existing ML systems [2, 11, 15–19] lies in the quantity regarding utilized models [2, 19], selection of optimization architecture [11, 15], duration of sample testing [16], evaluation of small dataset [11, 17], and/or randomized train/test splits [18, 19] to generate superior results. Though many studies published notable outcomes, more research focusing on a simplistic approach and faster training/testing of a model for patient-level detection with a larger image set needs firm development. Furthermore, in the remote and poverty-stricken areas, it will be difficult to find a dependable internet connection, high-performing computing source, and reliable power resource—though solar energy technology might be effective [20].

So, a desirable system needs the traits as below,

1. Research on patient-level detection by evaluating a large image set.
2. A compact method for IoT devices through a cloud platform.
3. Closer accuracy to NIH with less computation.

Microcontroller system gives ample scope for more development in this regard [21].

3 Proposed Methodology

Figure 1 depicts the proposed workflow to distinguish parasitized and uninfected cells.

3.1 Data Collection

The standard subset of the malaria dataset was collected from NIH, comprising 27,558 images [14]. The work was employed on equally distributed 13,779 parasitized and uninfected blood smear images after re-sizing those to 64×64 pixels for 3 (RGB) color mode channels. For each of the 2 class, about 85% of the total images were accounted for preparation and the last 15% was allocated for the experiment by data splits. Among the training dataset, around 10% of the whole dataset in each class was assigned for validation. Here, Fig. 2 represents a blood smear image subset provided by the NIH, whereas Table 1 illustrates the distribution of dataset for malaria detection consisting of training, validation, and testing for parasitized and uninfected categories.

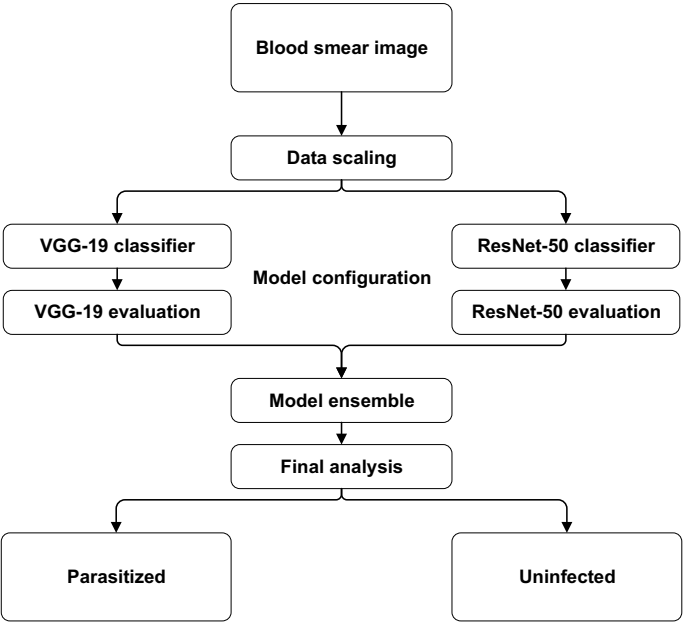


Fig. 1 Proposed workflow for malaria detection

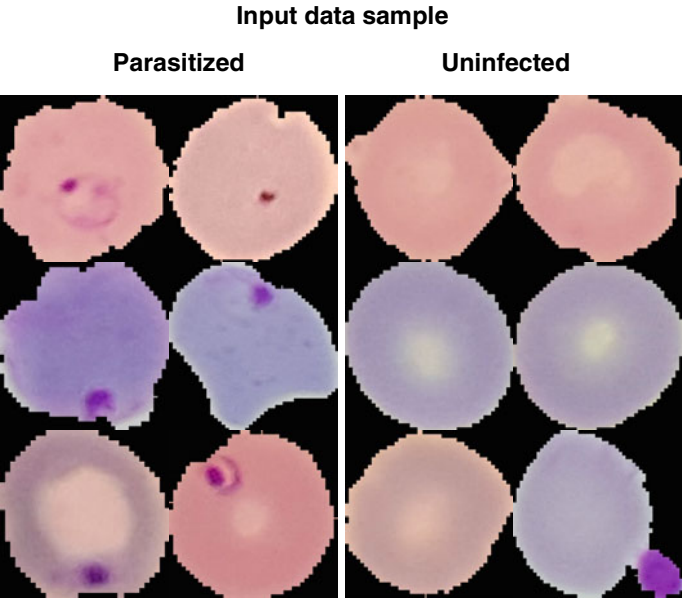


Fig. 2 A blood smear image subset provided by the NIH [14]

Table 1 Distribution of dataset for malaria detection

Dataset	Parasitized	Uninfected	Total
Training	10,301	10,301	20,602
Validation	1378	1378	2756
Testing	2100	2100	4200
Number of images	13,779	13,779	27,558

Bold indicates easier differentiation between total no. of images and distributed set of images

3.2 Model Configuration

For model training of both the VGG and ResNet architectures, the ImageNet dataset was used containing over 2200 object types and 15×10^6 high-resolution training images [22]. The VGG is a simple network architecture of 3×3 convolutional layers accumulated on another in progressing intensity that can capture little features skipping big patterns mostly. Else ways, ResNet is a complex network architecture, having the capability to discover large features as well as the small ones; whereas, residual module finally concatenates the mapping as element-wise addition and passes through a nonlinearity to generate output. In the case of both the models' training phase, predefined training data was fitted independently. Because deep neural network (DNN) has a considerable modeling potential, it may not be as productive on a small-sized dataset due to the parametric requirements of excessive data [23].

The implication of transfer learning (TL) decreased this issue because transferable weights can extract features [24]. As such, the transferable weights started the training process employing a pre-trained model. After collecting the pre-trained weights of ImageNet and initializing it to imply TL, the training process began. The first five, 17th, and 18th number layers (consisting of 4096 neurons each, then followed by a Softmax classifier) of VGG-19 were adjusted to improve the model. The weights of the first five layers were frozen to carry the transferable weights during the training.

To decrease the unnecessary computation of ImageNet, only 1024 neurons were chosen for both the 17th and 18th number layers after simulating a different number of neurons. By stochastic gradient descent (SGD) optimizer, the assembling of the model was executed with 0.0001 learning rate and 0.9 momentum. For matching the training data with the assembled model, the model was simulated many times with various epochs; and finally, for 100 epochs with 10 patience early stopping the best fit was generalized. The reference model for every architecture was selected regarding area under the receiver operating characteristic (AUROC) curvature with the highest value.

Figure 3 manifests the architectures of VGG-19 and ResNet-50 accordingly. The volume size reduction of VGG-19 is done by Maxpooling. For Resnet-50, $F(x)$ and $F'(x)$ are the residual mappings while x provides the identity mapping.

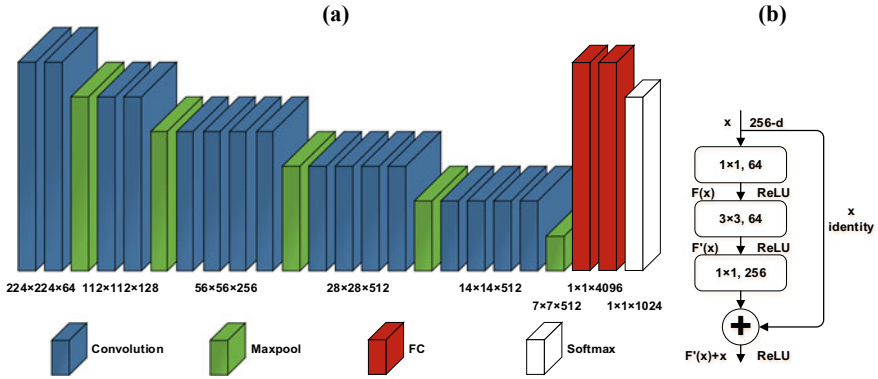


Fig. 3 Architecture of **a** VGG-19 and **b** ResNet-50 adapted from [25]

3.3 Model Ensemble

The ensemble method is an approach to construct a model by merging various architectures to generate a result with higher accuracy by devising sophisticated algorithms. It enriches both the existing dataset along with the model by continuous learning approach or combining diverse architectures. For getting a higher accuracy, the existing models are needed to be trained with better and accurate data as input for the weighted dataset. In this study, a weighted data manipulation technique was applied regarding a boosting ensemble model. So, we enhanced the weight of accurate data and calculated both the VGG-19 and ResNet-50 models' probability by applying Cohen's Kappa measurement. Then the weighted probability was determined by implying a formula through multiplying the model's probability with the weighted dataset. As such, the accuracy, precision, sensitivity rates generally increased in the ensemble model. The equations are specified as follows,

Consider, the weight of the n th classifier,

$$W_n = \frac{C_n}{\sum_{n=1}^i C_n} \quad (1)$$

And the prediction probability,

$$P = \sum_{n=1}^i P_n \times W_n \quad (2)$$

- Here, the n th classifier = C_n .
- And, the probability of the n th classifier = P_n .
- For the prediction probability >0.5 , the microscopic blood smear image will be classified as parasitized.

To evaluate the model performance by prediction, model testing was employed after the completion of model training.

3.4 Statistical Analysis

Exceptional outcomes for diversity were avoided by evaluating performance measuring metrics. For parametric evaluation of both real and speculated conditions, a confusion matrix was generated. Henceforth, the expressions are unfolded as specified,

- T_p = true-positive (determined accurately)
- T_n = true-negative (determined inaccurately)
- F_p = false-positive (undetermined accurately)
- F_n = false-negative (undetermined inaccurately).

The indices are expressed as stated,

$$\text{Accuracy} = \frac{T_p + T_n}{T_p + F_p + T_n + F_n} \quad (3)$$

$$\text{Precision} = \frac{T_p}{T_p + F_p} \quad (4)$$

$$\text{Sensitivity} = \frac{T_p}{T_p + F_n} \quad (5)$$

$$\text{Specifity} = \frac{T_n}{F_p + T_n} \quad (6)$$

$$F_\beta \text{ score} = \frac{(1 + \beta^2)PR}{\beta^2 P + R} \quad (7)$$

$$\text{Miss rate} = 1 - \frac{T_p}{T_p + F_n} \quad (8)$$

$$\text{Fall out} = 1 - \frac{T_n}{F_p + T_n} \quad (9)$$

$$\text{MCC} = \frac{(T_p \times T_n) - (F_p \times F_n)}{\sqrt{[(T_p + F_p)(T_p + F_n)(T_n + F_p)(T_n + F_n)]}} \quad (10)$$

$$\text{Cohen's Kappa} = \frac{\text{accuracy} - \text{expectation}}{1 - \text{expectation}} \quad (11)$$

The neural network model computation is a crucial portion of this analysis.

4 Experimental Results

After the implication of 2 different CNN architectures onto the microscopic blood smear image dataset, the performance measuring metrics was evaluated for each of the architectures. The best model was then chosen, whereas the investigation was ran using a moderately configured computer with Intel Core i7 2.5 GHz processor and 8 GB RAM. Table 2 presents the confusion matrix of different classifiers. The accurately determined value was comparatively higher in VGG-19 for the testing dataset. Among the 4200 instances for testing, 2051, 2004, and 2015 were determined accurately in VGG-19, ResNet-50, and ensemble models, respectively.

Figure 4 provides the ROC graph of VGG-19 and ResNet-50 correspondingly, where sensitivity plotted as a function of specificity. The AUROC for VGG-19 was 0.991, and AUROC of ResNet-50 was 0.987. Figure 5 exhibits the accuracy vs. epoch’s graphs of VGG-19 and ResNet-50 accordingly. The models did not over-learn the training datasets, whereas the trends almost reached towards the saturated levels—though the VGG-19 model could probably be trained a little more. For parallel processing, the TensorFlow graphics processing unit (GPU) was utilized as a backend library, reducing the model training and testing time.

Figure 6 reveals the performance metrics of malaria detection. From the subsequent graphical representations, it can be deduced that the ensemble model conveyed a better accuracy of 96.7% than both the VGG-19 and ResNet-50 (with 96.6% and 95.8% accuracy, respectively). The harmonic mean of the precision and recall (F_1 score) of the ensemble model is 0.967, where Matthews

Table 2 Confusion matrix of different classifiers

Classifiers	T_p	T_n	F_p	F_n
VGG-19	2051	2006	49	94
ResNet-50	2004	2018	96	82
Ensemble	2015	2047	53	85

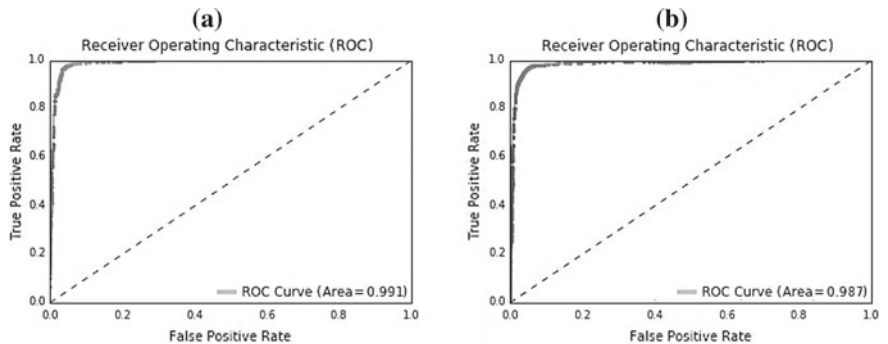


Fig. 4 ROC graph of a VGG-19 and b ResNet-50

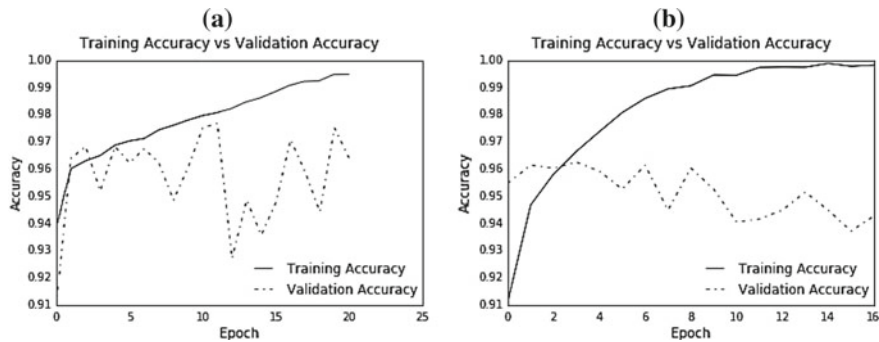


Fig. 5 Accuracy versus epochs graphs of **a** VGG-19 and **b** ResNet-50

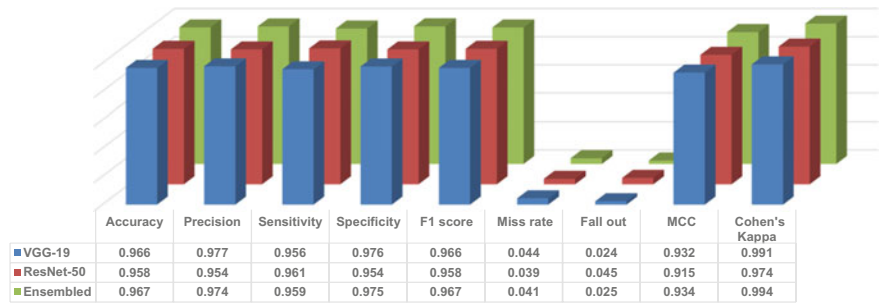


Fig. 6 Performance metrics of malaria detection

Correlation Coefficient (MCC) and Cohen’s Kappa showed a value of 0.934 and 0.994 accordingly. For each of these metrics (F_1 score, MCC, and Cohen’s Kappa), the ensemble model indicated a higher rate. On the other hand, for precision, specificity, and miss rate; VGG-19 speculated better marks; however, ResNet-50 expressed higher metrics in terms of sensitivity and fall out. Overall, the ensemble model functions comparatively better for malaria detection, which can still be improved with notable research.¹

Table 3 displays the performance differentiation of the stated architecture for malaria detection. Our proposed model showed an accuracy (0.967) close to the advanced method of Rajaraman et al. [15] for patient-level detection by exploring a different field. The model indicated a reasonable efficiency for almost all metrics compared to other methodological investigations (cell-level detection) as well, in terms of large dataset evaluation, simple model architecture, and faster training/testing process. For qualitative analysis, we upgraded the internal parameters by optimal solution, addressed the CNN architecture’s limitations by multiple modeling, and speculated the statistical evaluations by performance indices. In

¹Data supporting the conclusions of this research are accessible at github.com/ErtezaTawsif/Malaria.

Table 3 Performance comparison of proposed model for malaria detection

Methods	Accuracy	Sensitivity	Specificity	<i>F</i> -score
Our proposed model (patient-level)	0.967	0.960	0.975	0.967
Rajaraman et al. (patient-level) [15]	0.995	0.971	0.985	0.995
Bibin et al. [17]	0.963	0.976	0.959	0.897
Devi et al. [18]	0.963	0.930	0.968	0.853
Das et al. [11]	0.840	0.981	0.689	0.884

The bold metrics values indicate the performance measures of the presented architecture

summary, the proposed method conveyed noticeable performance with reliable outcomes. The final ensemble is done based on Cohen's Kappa statistics, combined with a boosting ensemble method. Our target was to give dynamic weights to the classifier, which holds better measures in Cohen's Kappa.

5 Conclusion

In the modern domain of medical technology, CADx proved to play a vital role regarding high speed and reliable efficiency to detect various diseases. Our proposed model thus explored a method to analyze the microscopic blood smear image dataset and form a classifier for detecting the contagious and often lethal parasitic disease, malaria. The system contains a classification method with a compact model to achieve a handful of benefits. Computing with a large set of images, our model performed with considerable accuracy within a short amount of time by a high-end GPU—recorded model training time was around 20 min and testing of a single study required less than 100 ms. The primary idea was to propagate this research to help medical personnel for taking precise and immediate therapeutic decisions considering poor, remote, and populous communities. Tuning parameters and applying different architectures could bring us a better result, thus the scope of improvement.

To improve the proposed model, recommendations can be made as below,

- Weight quantization on the respective models would allow a smaller version.
- The overall classification assignment should be available over the cloud.
- An end-to-end user-friendly mobile application needs to be developed.

The proposed model provides a design for automatic malaria detecting classifier via an application programming interface (API) with not only reasonable accuracy but also well-founded computation.

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References

1. Phillips, M.A., Burrows, J.N., Manyando, C., Van Huijsduijnen, R.H., Van Voorhis, W.C., Wells, T.N.C.: Malaria. *Nat. Rev. Dis. Primers* **3**, 17050 (2017). <https://doi.org/10.1038/nrdp.2017.50>
2. Rajaraman, S., Antani, S.K., Poostchi, M., Silamut, K., Hossain, M.A., Maude, R.J., Jaeger, S., Thoma, G.R.: Pre-trained convolutional neural networks as feature extractors toward improved malaria parasite detection in thin blood smear images. *PeerJ* (2018). <https://doi.org/10.7717/peerj.4568>
3. WHO: World Malaria Report 2016. WHO (2018)
4. Malaria Hero: A Web App for Faster Malaria Diagnosis. <https://blog.insightdatascience.com/blog-insightdatascience-com-malaria-hero-a47d3d5fc4bb>. Last accessed 30 June 2020
5. Deep Learning and Medical Image Analysis with Keras—PyImageSearch. <https://www.pyimagesearch.com/2018/12/03/deep-learning-and-medical-image-analysis-with-keras>. Last accessed 30 June 2020
6. Prevention, C.-C. for D.C. and: CDC—Malaria—About Malaria—Biology (2020)
7. WHO: How Malaria RDTs Work. WHO (2015)
8. Hommelshheim, C.M., Frantzeskakos, L., Huang, M., Ülker, B.: PCR amplification of repetitive DNA: a limitation to genome editing technologies and many other applications. *Sci. Rep.* **4**, 1–13 (2014). <https://doi.org/10.1038/srep05052>
9. Kido, S., Hirano, Y., Hashimoto, N.: Detection and classification of lung abnormalities by use of convolutional neural network (CNN) and regions with CNN features (R-CNN). In: 2018 International Workshop on Advanced Image Technology, IWAIT 2018, pp. 1–4. Institute of Electrical and Electronics Engineers Inc. (2018). <https://doi.org/10.1109/IWAIT.2018.8369798>
10. Ross, N.E., Pritchard, C.J., Rubin, D.M., Dusé, A.G.: Automated image processing method for the diagnosis and classification of malaria on thin blood smears. *Med. Biol. Eng. Comput.* **44**, 427–436 (2006). <https://doi.org/10.1007/s11517-006-0044-2>
11. Das, D.K., Ghosh, M., Pal, M., Maiti, A.K., Chakraborty, C.: Machine learning approach for automated screening of malaria parasite using light microscopic images. *Micron* **45**, 97–106 (2013). <https://doi.org/10.1016/j.micron.2012.11.002>
12. Poostchi, M., Silamut, K., Maude, R.J., Jaeger, S., Thoma, G.: Image Analysis and Machine Learning for Detecting Malaria (2018). <https://doi.org/10.1016/j.trsl.2017.12.004>
13. Lecun, Y., Bengio, Y., Hinton, G.: Deep Learning (2015). <https://www.nature.com/articles/nature14539>; <https://doi.org/10.1038/nature14539>
14. Malaria Datasets: National Library of Medicine. <https://lhncbc.nlm.nih.gov/publication/pub9932>. Last accessed 30 June 2020
15. Rajaraman, S., Jaeger, S., Antani, S.K.: Performance evaluation of deep neural ensembles toward malaria parasite detection in thin-blood smear images. *PeerJ* **7**, e6977 (2019). <https://doi.org/10.7717/peerj.6977>
16. Fuhad, K.M.F., Tuba, J.F., Sarker, M.R.A., Momen, S., Mohammed, N., Rahman, T.: Deep learning based automatic malaria parasite detection from blood smear and its smartphone based application. *Diagnostics* **10**, 329 (2020). <https://doi.org/10.3390/diagnostics10050329>
17. Bibin, D., Nair, M.S., Punitha, P.: Malaria parasite detection from peripheral blood smear images using deep belief networks. *IEEE Access* **5**, 9099–9108 (2017). <https://doi.org/10.1109/ACCESS.2017.2705642>
18. Devi, S.S., Roy, A., Singha, J., Sheikh, S.A., Laskar, R.H.: Malaria infected erythrocyte classification based on a hybrid classifier using microscopic images of thin blood smear. *Multimed. Tools Appl.* **77**, 631–660 (2018). <https://doi.org/10.1007/s11042-016-4264-7>
19. Gopakumar, G.P., Swetha, M., Sai Siva, G., Sai Subrahmanyam, G.R.K.: Convolutional neural network-based malaria diagnosis from focus stack of blood smear images acquired using custom-built slide scanner. *J. Biophotonics* **11**, e201700003 (2018). <https://doi.org/10.1002/jbio.201700003>

20. Efaz, E.T., Ava, A.A., Khan, M.T.A., Islam, M.M., Sultana, A.: Parametric analysis of CdTe/CdS thin film solar cell. *IJARCCCE* **5**, 401–404 (2016). <https://doi.org/10.17148/ijarcce.2016.5684>
21. Efaz, E.T., Mamun, A.Al, Salman, K., Kabir, F., Sakib, S.N., Khan, I.: Design of an indicative featured and speed controlled obstacle avoiding robot. In: 2019 International Conference on Sustainable Technologies for Industry 4.0, STI 2019. Institute of Electrical and Electronics Engineers Inc. (2019). <https://doi.org/10.1109/STI47673.2019.9068018>
22. Krizhevsky, A., Sutskever, I., Hinton, G.E.: ImageNet Classification with Deep Convolutional Neural Networks
23. Deng, J., Deng, J., Dong, W., Socher, R., Li, L., Li, K., Fei-fei, L.: Imagenet: a large-scale hierarchical image database. *CVPR* (2009)
24. Mondol, T.C., Iqbal, H., Hashem, M.M.A.: Deep CNN-based ensemble CADx model for musculoskeletal abnormality detection from radiographs. In: 2019 5th International Conference on Advances in Electrical Engineering, ICAEE 2019, pp. 392–397. Institute of Electrical and Electronics Engineers Inc. (2019). <https://doi.org/10.1109/ICAEE48663.2019.8975455>
25. Common Architectures in Convolutional Neural Networks. <https://www.jeremyjordan.me/convnet-architectures>. Last accessed 30 June 2020